CDAC CCEE

Data Collection & DBMS

MCQ Question Bank — 200+ Questions

Sections Covered	MySQL · MongoDB · SQL · NoSQL · DBMS Fundamentals · Normalization · PL/SQL · Transactions
Total Questions	200+ MCQs (with answers)
Difficulty	Beginner → Advanced (CCEE Level)
Pattern	Single correct answer · Conceptual + Query-based

SECTION 1: DBMS FUNDAMENTALS

Q1. Which type of database model uses a strict parent-child (tree) relationship only?

A) Hierarchical database
B) Network database
C) Object-oriented database
D) Relational database

✔ **Answer: A) Hierarchical database**

Note: Only ONE parent allowed per child node in hierarchical model.

Q2. A 'file' in DBMS is referred to as _____ in RDBMS.

A) Console
B) Schema
C) Table
D) Object

✔ **Answer: C) Table**

Q3. Which operator tests a column for the absence of data (NULL)?

A) ASSIGNMENT operator
B) LIKE operator
C) IS NULL operator
D) NOT operator

✔ **Answer: C) IS NULL operator**

Q4. The _____ refers to the way data is organized in and accessible from a DBMS.

A) Database hierarchy
B) Data organization
C) Data sharing
D) Data model

✔ **Answer: D) Data model**

Q5. Which is the most powerful language for working with RDBMS?

A) Embedded Programs
B) Dynamic Programs
C) Query Language
D) Static Language Programs

✔ **Answer: C) Query Language**

Q6. A table can have _____ primary key(s).

A) Zero
B) Only one
C) Two
D) Multiple

✔ **Answer: B) Only one**

Note: A table must have at most one PRIMARY KEY constraint, though it can span multiple columns (composite).

Q7. How many UNIQUE keys can a table have?

A) 1
B) 0
C) 2
D) Multiple

✔ **Answer: D) Multiple**

Q8. SQL stands for:

A) Structured Query Language
B) Solid Query Language
C) Simple Query Language

D) Science Query Language

✔ **Answer: A) Structured Query Language**

Q9. Which is more secure for data storage?

A) File System

B) RDBMS

C) Both equally

D) Neither

✔ **Answer: B) RDBMS**

Note: RDBMS provides role-based access control, authentication, and audit trails.

Q10. Excel is an example of:

A) DBMS

B) Database

C) RDBMS

D) ORDBMS

✔ **Answer: A) DBMS**

Note: Excel is a simple file-based DBMS, not a full RDBMS.

Q11. Oracle is an example of:

A) DBMS

B) Database

C) RDBMS

D) ORDBMS

✔ **Answer: D) ORDBMS**

Note: Oracle supports object-relational features.

Q12. Which of the following is an example of Non-Numerical Data?

A) Bank Balance

B) Examination Score

C) Employee Address

D) None of the above

✔ **Answer: C) Employee Address**

Q13. Which SQL keywords are supported?

A) INSERT

B) UPDATE

C) SELECT

D) All of the above

✔ **Answer: D) All of the above**

Q14. Which databases support SQL?

A) Oracle

B) SQL Server

C) MySQL

D) All of the above

✔ **Answer: D) All of the above**

Q15. A collection of data designed to be used by different people is called a/an:

A) Organization

B) Database

C) Relationship

D) Schema

✔ **Answer: B) Database**

Q16. Which term refers to the correctness and completeness of data in a database?

A) Data security

B) Data constraint

C) Data independence
D) Data integrity
✔ **Answer: D) Data integrity**

Q17. The relationship between DEPARTMENT and EMPLOYEE is:

A) One-to-one
B) One-to-many
C) Many-to-many
D) Many-to-one
✔ **Answer: B) One-to-many**

Note: One department can have many employees.

Q18. The number of rows in a table is called its:

A) Degree
B) Cardinality
C) Domain
D) Range
✔ **Answer: B) Cardinality**

Q19. The number of columns in a table is known as its:

A) Degree
B) Cardinality
C) Domain
D) Range
✔ **Answer: A) Degree**

Q20. In DBMS terms, ROWID is a:

A) Real column
B) Virtual column
C) Pseudo column
D) System column
✔ **Answer: C) Pseudo column**

Note: ROWID is an Oracle pseudo-column — not physically stored as a regular column.

Q21. Which component holds the definitions of all data tables?

A) Database
B) Data dictionary
C) Data source
D) Data mining
✔ **Answer: B) Data dictionary**

Q22. If a database state no longer reflects the real world, it is called:

A) Consistent state
B) Parallel state
C) Durable state
D) Inconsistent state
✔ **Answer: D) Inconsistent state**

Q23. Each modification done in a database transaction is first recorded into the:

A) Hard drive
B) Log
C) Disk
D) Datamart
✔ **Answer: B) Log**

Q24. Overall logical structure of a database can be expressed graphically by:

A) ER diagram
B) Records

C) Relations
D) Hierarchy
✔ **Answer: A) ER diagram**

Q25. Specialization is a _____ relationship; Composition is a _____ relationship.

A) HAS-A / IS-A
B) IS-A / HAS-A
C) Part-of / Is-a
D) None of the above
✔ **Answer: B) IS-A / HAS-A**

SECTION 2: SQL — DDL, DML, DCL, TCL

2.1 DDL — Data Definition Language

Q26. Which SQL statement is used to define the structure of a relation?

A) DML
B) DDL
C) Query
D) Relational Schema

✔ **Answer: B) DDL (Data Definition Language)**

Q27. CREATE TABLE employee (name VARCHAR, id INTEGER) — this is a _____ statement.

A) DML
B) DDL
C) View
D) Integrity constraint

✔ **Answer: B) DDL**

Q28. To remove a relation from an SQL database, use the _____ command.

A) DELETE
B) PURGE
C) REMOVE
D) DROP TABLE

✔ **Answer: D) DROP TABLE**

Q29. Which command deletes table data AND the table structure?

A) TRUNCATE
B) DELETE
C) ALTER
D) DROP

✔ **Answer: D) DROP**

Note: DROP removes structure + data + indexes. TRUNCATE removes only data but keeps structure.

Q30. Which command removes all rows but keeps the table structure?

A) TRUNCATE
B) DROP
C) DELETE
D) ALTER

✔ **Answer: A) TRUNCATE**

Note: TRUNCATE is DDL — auto-committed and faster than DELETE.

Q31. Which command is used to add a column to an existing table?

A) CREATE
B) INSERT
C) ALTER TABLE ... ADD COLUMN
D) UPDATE

✔ **Answer: C) ALTER TABLE ... ADD COLUMN**

Q32. To include an integrity constraint in an existing relation, use:

A) CREATE TABLE
B) MODIFY TABLE
C) ALTER TABLE
D) DROP TABLE

✔ **Answer: C) ALTER TABLE**

Q33. Which command creates a database named 'student'?

A) CREATE ?I student
B) CREATE DATABASE student
C) DATABASE /student
D) DATABSE student

✔ **Answer: B) CREATE DATABASE student**

Q34. The correct syntax to drop a constraint is:

A) ALTER TABLE t DROP CONSTRAINT cname;
B) ALTER TABLE t DELETE CONSTRAINT cname;
C) ALTER TABLE t DROP cname;
D) None of the above
✔ **Answer: A) ALTER TABLE t DROP CONSTRAINT cname;**

Q35. TRUNCATE TABLE statement:

A) Drops a table temporarily
B) Removes all rows of a table
C) Removes specified number of rows
D) Removes all constraints
✔ **Answer: B) Removes all rows of a table**

2.2 DML — Data Manipulation Language

Q36. Which SQL provides the ability to INSERT, DELETE, UPDATE, and QUERY the database?

A) DML
B) DDL
C) Query
D) Relational Schema
✔ **Answer: A) DML (Data Manipulation Language)**

Q37. Which of the following can add a row to a table?

A) ADD
B) INSERT
C) UPDATE
D) ALTER
✔ **Answer: B) INSERT**

Q38. Select the correct INSERT statement:

A) INSERT VALUES ('15','Hari',45) INTO stdinfo;
B) INSERT VALUES INTO stdinfo ('15','Hari',45);
C) INSERT stdinfo VALUES ('15','Hari',45);
D) INSERT INTO stdinfo VALUES ('15','Hari',45);
✔ **Answer: D) INSERT INTO stdinfo VALUES ('15','Hari',45);**

Q39. Which query deletes data from a table?

A) DROP
B) DELETE
C) UPDATE
D) ALTER
✔ **Answer: B) DELETE**

Q40. How to delete records from studentinfo where sname='Hari Prasad'?

A) DELETE FROM TABLE studentinfo WHERE sname='Hari Prasad';
B) DELETE FROM studentinfo WHERE sname='Hari Prasad';
C) DELETE FROM studentinfo WHERE COLUMN sname='Hari Prasad';
D) DELETE FROM studentinfo WHERE sname LIKE 'Hari Prasad';
✔ **Answer: B) DELETE FROM studentinfo WHERE sname='Hari Prasad';**

Q41. The command to remove rows from table CUSTOMER is:

A) REMOVE FROM CUSTOMER ...
B) DROP FROM CUSTOMER ...
C) DELETE FROM CUSTOMER WHERE ...
D) UPDATE FROM CUSTOMER ...
✔ **Answer: C) DELETE FROM CUSTOMER WHERE ...**

Q42. Omitting WHERE from a DELETE statement results in:

A) Error — no records to delete
B) Prompts for criteria
C) Syntax error
D) Deletes ALL records
✔ **Answer: D) Deletes ALL records from the table**

Q43. Which query updates instructor salary by 5%?

A) UPDATE instructor WHERE salary = salary * 1.05;
B) UPDATE instructor SET salary = salary * 1.05;
C) ALTER instructor SET salary = salary * 1.05;
D) MODIFY instructor salary = salary * 1.05;
✔ **Answer: B) UPDATE instructor SET salary = salary * 1.05;**

Q44. The keyword used in UPDATE to specify columns to change is:

A) WHERE
B) SET
C) IN
D) SELECT
✔ **Answer: B) SET**

Q45. Which DML command is used for inserting rows in tables?

A) DDL
B) DML
C) DCL
D) DQL
✔ **Answer: B) DML**

2.3 SELECT Queries

Q46. The correct order of keywords for SQL SELECT statements is:

A) SELECT, FROM, WHERE
B) FROM, WHERE, SELECT
C) WHERE, FROM, SELECT
D) SELECT, WHERE, FROM
✔ **Answer: A) SELECT, FROM, WHERE**

Q47. A SELECT command without a WHERE clause returns:

A) Records matching previous WHERE
B) All records from a table
C) SELECT is invalid without WHERE
D) Nothing
✔ **Answer: B) All the records from a table**

Q48. The SQL WHERE clause:

A) Limits column data returned
B) Limits ROW data returned
C) Both A and B
D) Neither A nor B
✔ **Answer: B) Limits the ROW data returned**

Q49. Which SQL clause restricts the rows returned by a query?

A) AND
B) WHERE
C) GROUP
D) FROM
✔ **Answer: B) WHERE**

Q50. The result of a SELECT statement can contain duplicate rows.

A) False
B) True
C) Only with DISTINCT
D) Only with UNION
✔ **Answer: B) True**

Note: Use SELECT DISTINCT to eliminate duplicates.

Q51. To select only one copy of each set of duplicate rows:

A) SELECT DISTINCT
B) SELECT UNIQUE
C) SELECT DIFFERENT
D) All of the above
✔ **Answer: A) SELECT DISTINCT**

Q52. Which SELECT statement retrieves all unique course names from studentinfo?

A) SELECT DISTINCT coursename FROM studentinfo;
B) SELECT UNIQUE coursename FROM studentinfo;
C) SELECT DISTINCT coursename FROM TABLE studentinfo;
D) SELECT INDIVIDUAL coursename FROM studentinfo;
✔ **Answer: A) SELECT DISTINCT coursename FROM studentinfo;**

Q53. Which SQL query is CORRECT?

A) Select * from emp where eid=100 , ename='a'
B) Select * from emp where eid=100 ename='a'
C) Select * where eid=100 and ename='a' from emp
D) Select * from emp where eid=100 and ename='a'
✔ **Answer: D) Select * from emp where eid=100 and ename='a'**

Q54. A subquery in an SQL SELECT statement is enclosed in:

A) Braces { }
B) CAPITAL LETTERS
C) Parentheses ( )
D) Brackets [ ]
✔ **Answer: C) Parentheses ( )**

Q55. Which query sorts data from empinfo in ascending order?

A) SELECT * FROM empinfo ORDER BY age;
B) SELECT * FROM empinfo ORDER age;
C) SELECT * FROM empinfo ORDER BY COLUMN age;
D) SELECT * FROM empinfo SORT BY age;
✔ **Answer: A) SELECT * FROM empinfo ORDER BY age;**

Q56. Default sort order in MySQL ORDER BY clause is:

A) Ascending
B) Descending
C) No sort
D) Both
✔ **Answer: A) Ascending**

Q57. A table may be joined to itself (self-join).

A) True
B) False
C) Only with aliases
D) Only in Oracle
✔ **Answer: A) True**

Q58. SELECT * from emp where salary is not null — selects:

A) Tuples with null salary
B) Tuples with NO null values
C) Tuples with any salary
D) All tuples
✔ **Answer: B) Tuples with no null values**

Q59. Which query selects data starting with letter 'r'?

A) WHERE sname LIKE 'r%'
B) WHERE sname LIKE '%r%'
C) WHERE sname LIKE '%r'
D) WHERE sname LIKE '_r%'
✔ **Answer: A) WHERE sname LIKE 'r%'**

Q60. In a LIKE clause, to match any value ending in 'qpt':

A) LIKE %qpt
B) LIKE *qpt
C) LIKE qpt$
D) LIKE ^.*qpt$
✔ **Answer: A) LIKE %qpt**

Q61. In a LIKE clause, to match any 6-letter value:

A) LIKE ??????
B) LIKE .{6}
C) LIKE .....
D) LIKE _____ (six underscores)
✔ **Answer: D) LIKE _____ (six underscore characters)**

Note: In SQL, _ matches exactly ONE character; % matches zero or more.

Q62. The wildcard % used in SQL strings means:

A) Any number of characters

B) One or more characters
C) At least one character
D) At least one alphabet
✔ **Answer: A) Any number of characters**

Q63. The correct pattern-matching sign for a SINGLE character in SQL is:

A) _
B) ?
C) *
D) %
✔ **Answer: A) Underscore (_)**

2.4 Aggregate Functions & GROUP BY

Q64. Which of the following is an aggregate function in SQL?

A) UNION
B) LIKE
C) GROUP BY
D) MAX
✔ **Answer: D) MAX**

Q65. The five built-in aggregate functions provided by SQL are:

A) COUNT, SUM, AVG, MAX, MIN
B) SUM, AVG, MIN, MAX, MULT
C) SUM, AVG, MULT, DIV, MIN
D) SUM, AVG, MIN, MAX, NAME
✔ **Answer: A) COUNT, SUM, AVG, MAX, MIN**

Q66. Aggregate functions take _____ as input and return a single value.

A) Single value
B) Collection of values
C) Aggregate value
D) Both A & B
✔ **Answer: B) Collection of values**

Q67. Which function finds the mean (average) salary?

A) MEAN(salary)
B) AVG(salary)
C) SUM(salary)
D) COUNT(salary)
✔ **Answer: B) AVG(salary)**

Q68. All aggregate functions EXCEPT _____ ignore NULL values.

A) COUNT(attribute)
B) COUNT(*)
C) AVG
D) SUM
✔ **Answer: B) COUNT(*)**

Note: COUNT(*) counts ALL rows including NULLs; COUNT(col) ignores NULLs.

Q69. Which of the following is NOT a valid aggregate function?

A) COUNT
B) MIN
C) MAX
D) COMPUTE
✔ **Answer: D) COMPUTE**

Q70. Which of the following is NOT a GROUP function?

A) AVG()
B) SQRT()
C) SUM()
D) MAX()
✔ **Answer: B) SQRT()**

Q71. Can we have a HAVING clause without GROUP BY?

A) Yes
B) No
C) Only in Oracle
D) Only with aggregate functions
✔ **Answer: B) No**

Note: HAVING is always used with GROUP BY to filter group results.

Q72. The correct order of precedence of SQL clauses is:

A) HAVING, GROUP BY, WHERE
B) WHERE, GROUP BY, HAVING
C) GROUP BY, HAVING, WHERE
D) HAVING, WHERE, GROUP BY
✔ **Answer: B) WHERE, GROUP BY, HAVING**

Q73. In a GROUP BY query, SELECT columns must:

A) Be aggregate functions only
B) Also appear in GROUP BY clause
C) Appear in GROUP BY and HAVING
D) Both A and B
✔ **Answer: D) Both A and B**

Q74. The correct HAVING clause syntax order is:

A) SELECT col FROM tbl HAVING cond GROUP BY col;
B) SELECT col FROM tbl GROUP BY col HAVING cond;
C) SELECT col HAVING cond FROM tbl GROUP BY col;
D) SELECT col HAVING cond GROUP BY col FROM tbl;
✔ **Answer: B) SELECT col FROM tbl GROUP BY col HAVING cond;**

2.5 Joins

Q75. Which join returns the HIGHEST number of rows?

A) Cross Join
B) Left Join
C) Inner Join
D) Right Join
✔ **Answer: A) Cross Join**

Note: Cross Join = Cartesian product of both tables.

Q76. Which join does NOT preserve non-matched tuples?

A) Left outer join
B) Right outer join
C) Inner join
D) Natural join
✔ **Answer: C) Inner join**

Q77. What type of join includes rows WITHOUT matching values?

A) Equi-join
B) Natural join
C) Outer join
D) All of the mentioned
✔ **Answer: C) Outer join**

Q78. How many tables can be included in a JOIN?

A) One
B) Two
C) Three
D) All of the mentioned
✔ **Answer: D) All of the mentioned**

Q79. SELECT * FROM R, S is equivalent to:

A) SELECT * FROM R NATURAL JOIN S
B) SELECT * FROM R CROSS JOIN S
C) SELECT * FROM R UNION JOIN S
D) SELECT * FROM R INNER JOIN S
✔ **Answer: B) SELECT * FROM R CROSS JOIN S**

Q80. Which join type is NOT supported by MySQL?

A) FULL OUTER JOIN
B) LEFT JOIN
C) RIGHT JOIN
D) INNER JOIN
✔ **Answer: A) FULL OUTER JOIN**

Note: MySQL does not support FULL OUTER JOIN directly; use UNION of LEFT and RIGHT joins.

Q81. The ON condition in a JOIN allows:

A) General predicate over joined relations
B) Only equality conditions
C) Only inequality conditions
D) None of the above
✔ **Answer: A) General predicate over the relations being joined**

Q82. A Cartesian product is produced when:

A) A GROUP function is used
B) A join SELECT has no WHERE clause
C) Fuzzy logic is applied
D) A subquery is nested
✔ **Answer: B) A join SELECT has no WHERE/ON clause**

2.6 DCL & TCL

Q83. Which command is used to get back privileges offered by GRANT?

A) GRANT
B) REVOKE
C) EXECUTE
D) RUN
✔ **Answer: B) REVOKE**

Q84. GRANT and REVOKE are _____ commands.

A) DDL
B) DML
C) DCL
D) DQL
✔ **Answer: C) DCL (Data Control Language)**

Q85. Which component of SQL is used for granting and revoking privileges?

A) DDL
B) DML
C) DCL
D) DQL
✔ **Answer: C) DCL**

Q86. A _____ consists of a sequence of query and/or update statements.

A) Transaction
B) Commit
C) Rollback
D) Flashback
✔ **Answer: A) Transaction**

Q87. Which of the following makes a transaction PERMANENT in the database?

A) View
B) Commit
C) Rollback
D) Flashback
✔ **Answer: B) Commit**

Q88. To UNDO work done after the last commit:

A) View
B) Commit
C) Rollback
D) Flashback
✔ **Answer: C) Rollback**

Q89. In case of shutdown during transaction before commit, which is done automatically?

A) View
B) Commit
C) Rollback
D) Flashback
✔ **Answer: C) Rollback**

Q90. Which of the following is an atomic sequence of database actions?

A) Transaction
B) Concurrency
C) Relations
D) All of the mentioned
✔ **Answer: A) Transaction**

Q91. _____ means data used during a transaction cannot be used by another transaction until the first completes.

A) Serializability

B) Atomicity
C) Isolation
D) Time stamping
✔ **Answer: C) Isolation**

Q92. Rollback and Commit affect:

A) Only DML statements
B) Only DDL statements
C) Both DML and DDL
D) All SQL*Plus statements
✔ **Answer: A) Only DML statements**

Note: DDL statements (CREATE, DROP, ALTER) are auto-committed.

Q93. Which isolation level is the DEFAULT in MySQL?

A) Read Uncommitted
B) Read Committed
C) Repeatable Reads
D) Serializable
✔ **Answer: C) Repeatable Reads**

Q94. Which isolation level allows DIRTY READS?

A) Repeatable Reads
B) Read Committed
C) Read Uncommitted
D) Serializable
✔ **Answer: C) Read Uncommitted**

Q95. Which isolation level does NOT allow PHANTOM READS?

A) Repeatable Reads
B) Read Committed
C) Read Uncommitted
D) Serializable
✔ **Answer: D) Serializable**

Q96. Which is the STRICTEST isolation level?

A) Read Uncommitted
B) Read Committed
C) Repeatable Reads
D) Serializable
✔ **Answer: D) Serializable**

Q97. Which isolation level does NOT allow NON-REPEATABLE READS?

A) Repeatable Reads
B) Read Committed
C) Both A and B
D) None
✔ **Answer: A) Repeatable Reads**

Q98. Which command is used to start a transaction?

A) START TRANSACTION
B) TRANSACTION
C) START TRAN
D) BEGIN WORK
✔ **Answer: A) START TRANSACTION**

Q99. MySQL runs with _____ mode enabled by default.

A) COMMIT
B) AUTOCOMMIT

C) ROLLBACK
D) SAVE POINT
✔ **Answer: B) AUTOCOMMIT**

Q100. The WITH GRANT OPTION clause of the GRANT command allows the recipient to:

A) Own the granted object
B) Pass the privilege to others
C) Revoke the privilege
D) None of the above
✔ **Answer: B) Pass the privilege to others**

SECTION 3: KEYS, CONSTRAINTS & NORMALIZATION

3.1 Keys & Constraints

Q101. The primary key must be:

A) Unique only
B) Not NULL only
C) Both Unique and Not NULL
D) Either Unique or Not NULL
✔ **Answer: C) Both Unique and Not NULL**

Q102. How many NULL values can a PRIMARY KEY column have in MySQL?

A) Multiple
B) 0
C) 1
D) 2
✔ **Answer: B) 0 — Primary key cannot have NULL values**

Q103. How many NULL values can a UNIQUE KEY column have in MySQL?

A) 0
B) 1
C) 2
D) Multiple
✔ **Answer: D) Multiple**

Note: UNIQUE allows multiple NULLs in MySQL because NULL != NULL.

Q104. Can a child table's FOREIGN KEY column have NULL values?

A) True
B) False
C) Depends on DB
D) Only if CHECK constraint allows
✔ **Answer: A) True**

Note: Foreign key columns can be NULL unless NOT NULL is specified.

Q105. Does MySQL support COMPOSITE Primary Keys?

A) True
B) False
C) Only in InnoDB
D) Only in MyISAM
✔ **Answer: A) True**

Q106. Does MySQL support COMPOSITE Unique Keys?

A) True
B) False
C) Only in v8.0+
D) Only with InnoDB
✔ **Answer: A) True**

Q107. Does MySQL support COMPOSITE Foreign Keys?

A) True
B) False
C) Only in Oracle
D) Only with triggers
✔ **Answer: A) True**

Q108. Composite key is made up of:

A) One column
B) One super key
C) One foreign key
D) Two or more columns

✔ **Answer: D) Two or more columns**

Q109. Foreign key is the one in which the _____ of one relation is referenced in another.

A) Foreign key
B) Primary key
C) References
D) Check constraint
✔ **Answer: B) Primary key**

Q110. Referential integrity constraint can be specified with the help of:

A) Primary key
B) Foreign key
C) Secondary key
D) None of the above
✔ **Answer: B) Foreign key**

Q111. Which of the following is NOT an integrity constraint?

A) NOT NULL
B) POSITIVE
C) UNIQUE
D) CHECK 'predicate'
✔ **Answer: B) POSITIVE**

Note: POSITIVE is not a standard SQL constraint.

Q112. Which integrity constraint enforces relationship between two tables?

A) Primary key
B) CHECK
C) REFERENCES (Foreign key)
D) None
✔ **Answer: C) REFERENCES (Foreign key)**

Q113. How many Auto Increment columns can a table have?

A) 0
B) 1
C) Both 0 or 1
D) Multiple
✔ **Answer: C) Both 0 or 1 (at most one)**

Q114. On-DELETE-CASCADE option with foreign key does:

A) Delete from master table only
B) Delete from transaction table only
C) Removes all transaction entries along with master
D) None
✔ **Answer: C) Removes all transaction entries along with master**

Q115. Which constraint enforces Entity Integrity in RDBMS?

A) Primary key
B) NOT NULL
C) UNIQUE
D) Foreign key
✔ **Answer: A) Primary key**

3.2 Normalization

Q116. A relation is in _____ if every field contains only atomic values (no lists or sets).

A) 1NF
B) 2NF
C) 3NF
D) BCNF

✔ **Answer: A) 1NF — First Normal Form**

Q117. A function that has NO partial dependencies is in:

A) 3NF
B) 2NF
C) 4NF
D) BCNF

✔ **Answer: B) 2NF**

Note: 2NF: 1NF + no partial dependency on composite primary key.

Q118. If every non-key attribute is functionally dependent on the ENTIRE primary key, the relation is in:

A) 1NF
B) 2NF
C) 3NF
D) 4NF

✔ **Answer: B) 2NF**

Q119. Third Normal Form (3NF) is based on the concept of:

A) Closure dependency
B) Transitive dependency
C) Normal dependency
D) Functional dependency

✔ **Answer: B) Transitive Dependency**

Note: 3NF: 2NF + no transitive dependency of non-key on primary key.

Q120. Every Boyce-Codd Normal Form (BCNF) is in:

A) 1NF
B) 3NF
C) 2NF
D) All of the above

✔ **Answer: D) All of the above — BCNF ⊃ 3NF ⊃ 2NF ⊃ 1NF**

Q121. 4NF is designed to cope with:

A) Transitive dependency
B) Join dependency
C) Multi-valued dependency
D) None of these

✔ **Answer: C) Multi-valued dependency**

Q122. Which Normal Form allows dependency of one non-key attribute on another non-key attribute?

A) 2NF
B) 3NF
C) 4NF
D) None

✔ **Answer: A) 2nd Normal Form**

Note: 3NF eliminates transitive dependency; 2NF still allows it.

Q123. What level of Normalization is present in STAR Schema?

A) 2NF
B) 3NF
C) 4NF
D) 1NF

✔ **Answer: A) 2nd Normal Form**

Q124. What level of Normalization is present in SNOWFLAKE Schema?

A) 2NF
B) 3NF
C) 4NF
D) 1NF

✔ **Answer: B) 3rd Normal Form**

Q125. Normalization is needed to reduce data:

A) Redundancy
B) Complexity
C) Visualization
D) Processing

✔ **Answer: A) Redundancy**

Q126. Denormalization is needed to improve _____ efficiency.

A) SELECT
B) INSERT
C) UPDATE
D) DELETE

✔ **Answer: A) SELECT**

Note: Denormalization trades storage redundancy for faster read performance.

Q127. Repeating groups are removed in which Normal Form?

A) 1NF
B) 2NF
C) 3NF
D) 4NF

✔ **Answer: A) 1NF**

Q128. A relation is in this form if it is in BCNF and has no multi-valued dependencies:

A) 2NF
B) 3NF
C) 4NF
D) None of the above

✔ **Answer: C) 4NF**

Q129. 2NF applies only to tables with:

A) Single primary key
B) Composite primary keys
C) Foreign keys
D) Unique keys

✔ **Answer: B) Composite primary keys**

SECTION 4: MySQL — ADVANCED

4.1 MySQL Commands & Queries

Q130. Which set operator is supported by MySQL?

A) UNION
B) INTERSECT
C) MINUS
D) EXCEPT
✔ **Answer: A) UNION**

Note: MySQL supports UNION and UNION ALL. INTERSECT and MINUS/EXCEPT added in MySQL 8.0.

Q131. Does MySQL support CASE statements?

A) True
B) False
C) Only in stored procedures
D) Only in triggers
✔ **Answer: A) True**

Q132. Which is NOT a valid aggregate function in MySQL?

A) AVG
B) SUM
C) WITH
D) MIN
✔ **Answer: C) WITH**

Note: WITH is used for CTEs (Common Table Expressions), not an aggregate function.

Q133. The EXISTS keyword is TRUE if:

A) Any row in the subquery meets the condition
B) All rows in the subquery fail the condition
C) Both conditions are met
D) Neither condition is met
✔ **Answer: A) Any row in the subquery meets the condition**

Q134. We test for NONEXISTENCE of tuples in a subquery using:

A) NOT EXIST
B) NOT EXISTS
C) EXISTS
D) EXIST
✔ **Answer: B) NOT EXISTS**

Q135. The phrase 'greater than at least one' is represented in SQL by:

A) < ALL
B) < SOME
C) > ALL
D) > ANY
✔ **Answer: D) > ANY (or > SOME)**

Q136. Which SQL command is used to select the currently connected user in MySQL?

A) SELECT user FROM mysql.user
B) SELECT user
C) SHOW USERS
D) SELECT current_user()
✔ **Answer: D) SELECT current_user()**

Q137. Which command shows all users in MySQL?

A) SELECT user FROM mysql.user
B) SELECT user
C) SHOW USERS
D) SELECT current_user()

✔ **Answer: A) SELECT user FROM mysql.user**

Q138. What is used to FETCH data row by row from a table?

A) Cursor
B) Variable
C) Both
D) None
✔ **Answer: A) Cursor**

Q139. What is used to OPTIMIZE search performance?

A) View
B) Group By Clause
C) Index
D) Procedure
✔ **Answer: C) Index**

Q140. Which of the following can be called INSIDE queries?

A) Procedures
B) Functions
C) Triggers
D) All
✔ **Answer: B) Functions**

Note: Only Functions can be called within SQL queries (SELECT, WHERE, etc.).

Q141. Which CANNOT be executed by users directly?

A) Triggers
B) Functions
C) Procedures
D) None
✔ **Answer: A) Triggers**

Note: Triggers fire automatically on DML events, not called directly by users.

Q142. The BETWEEN operator is used to test whether a value lies within a range.

A) True
B) False
C) Only with numbers
D) Only with dates
✔ **Answer: A) True**

Q143. Which query is correct for finding salaries that are NULL?

A) SELECT salary FROM emp WHERE salary=null
B) SELECT salary FROM emp WHERE salary='null'
C) SELECT salary FROM emp WHERE salary IS NOT NULL
D) SELECT salary FROM emp WHERE salary IS NULL
✔ **Answer: D) SELECT salary FROM emp WHERE salary IS NULL**

Q144. A NULL value is treated as:

A) Blank
B) 0
C) Unknown/absent
D) All of the above
✔ **Answer: C) Unknown/absent value**

Note: NULL is not blank or 0; it represents unknown or missing data.

Q145. Any value added to NULL gives:

A) The value itself
B) 0
C) NULL

D) Unpredictable

✔ **Answer: C) NULL**

Note: Any arithmetic operation with NULL results in NULL.

Q146. Which keys can have NULL values?

A) Primary Key
B) Unique Key
C) Foreign Key
D) Both Unique Key and Foreign Key

✔ **Answer: D) Both Unique Key and Foreign Key**

Q147. The USE command in MySQL should be used to:

A) Load code from another file
B) Avoid for security reasons
C) Synonym for SELECT
D) Choose the database to use after connecting

✔ **Answer: D) Choose the database you want to use once connected to MySQL**

Q148. Which query correctly uses comparison operators?

A) SELECT sname FROM studentinfo WHERE age>50 and <80;
B) SELECT sname FROM studentinfo WHERE age>50 and age<80;
C) SELECT sname FROM studentinfo WHERE age>50 and WHERE age<80;
D) None of the above

✔ **Answer: B) SELECT sname FROM studentinfo WHERE age>50 and age<80;**

Q149. Which of the following is an INVALID table name in Oracle?

A) EMP_2008
B) Emp_2008
C) emp_2008
D) 2008_EMP

✔ **Answer: D) 2008_EMP**

Note: Table names must begin with a letter in Oracle.

Q150. In Oracle, the maximum number of columns per table is:

A) 254
B) 255
C) 256
D) 1000

✔ **Answer: D) 1000**

4.2 MySQL Functions

Q151. The character used for CONCATENATING strings in Oracle is:

A) &&
B) +
C) ||
D) None
✔ **Answer: C) || (double pipe)**

Q152. INITCAP('cdac') returns:

A) cdac
B) CDAC
C) Cdac
D) cdAC
✔ **Answer: C) Cdac**

Q153. SELECT SUBSTR('CHARRIS', -3) FROM dual gives:

A) RIS
B) CHA
C) ARRIS
D) ARR
✔ **Answer: A) RIS**

Note: Negative index counts from end; last 3 chars of CHARRIS = RIS.

Q154. The value of INSTR('CALIFORNIA', 'A', 1, 2) is:

A) 2
B) 1
C) 10
D) 9
✔ **Answer: C) 10**

Note: Second occurrence of 'A' in CALIFORNIA is at position 10.

Q155. Which function makes a string of certain length by padding characters on the LEFT?

A) LTRIM
B) LPAD
C) SUBSTR
D) INSTR
✔ **Answer: B) LPAD**

Q156. The DUAL table has:

A) One row, many columns
B) One column and many rows
C) One row and ONE column
D) Many rows and many columns
✔ **Answer: C) One row and ONE column (named DUMMY)**

Q157. The NVL() function:

A) Distributes output across columns
B) Returns alternate for non-NULL values
C) Returns alternate output for NULL values
D) Nullifies column values
✔ **Answer: C) Allows you to specify alternate output for NULL column values**

Q158. MONTHS_BETWEEN returns:

A) A number
B) A character
C) A date
D) A month
✔ **Answer: A) A number**

Q159. The SQL query to retrieve CURRENT TIME in Oracle is:

A) SELECT SysTime FROM dual;
B) SELECT SysDate FROM dual;
C) SELECT TO_CHAR(SysDate,'HH:MI:SS') FROM dual;
D) SELECT TO_CHAR(SysDate,'HH:MM:SS') FROM dual;
✔ **Answer: C) SELECT TO_CHAR(SysDate,'HH:MI:SS') FROM dual;**

Q160. Which function is similar to IF-THEN-ELSE in SQL?

A) SQRT()
B) DECODE()
C) NEW_TIME()
D) ROWIDTOCHAR()
✔ **Answer: B) DECODE()**

Q161. The CHR function:

A) Accepts a number & returns the character
B) Accepts a character & returns the number
C) Accepts a number & returns a number
D) None of the above
✔ **Answer: A) Accepts a number and returns the character**

Q162. SELECT CEIL(MOD(POWER(2,3), FLOOR(3.5))) FROM dual; Result:

A) 0
B) 1
C) 2
D) 4
✔ **Answer: C) 2**

Note: POWER(2,3)=8; FLOOR(3.5)=3; MOD(8,3)=2; CEIL(2)=2

Q163. Which of the following is NOT a number function?

A) SINH()
B) TO_NUMBER()
C) SQRT()
D) ROUND()
✔ **Answer: B) TO_NUMBER()**

Note: TO_NUMBER() is a conversion function, not a number/math function.

SECTION 5: NoSQL DATABASES

5.1 NoSQL Concepts

Q164. NoSQL databases are most often referred to as:

A) Object oriented
B) Relational
C) Distributed
D) All of the above
✔ **Answer: D) All of the above**

Q165. NoSQL database is used mainly for handling large volumes of _____ data.

A) Structured
B) Semi-structured
C) Unstructured
D) All of the above
✔ **Answer: D) All of the above**

Q166. The NoSQL databases can be categorized into:

A) Column oriented
B) Document oriented
C) Key-value and graph oriented
D) All of the above
✔ **Answer: D) All of the above**

Q167. Which of the following is NOT a feature of NoSQL databases?

A) Data can be easily held across multiple servers
B) Relational data
C) Scalability
D) All of the above
✔ **Answer: B) Relational data**

Note: NoSQL excels with non-relational, schema-free data.

Q168. Most NoSQL databases support automatic _____ for high availability.

A) Processing
B) Availability
C) Replication
D) All of the options
✔ **Answer: D) All of the options**

Q169. Which of the following is FALSE with respect to relational DBMS?

A) Highly normalized tables may require complex joins
B) Foreign key constraints are referential integrity constraints
C) Primary key uniquely identifies each row
D) NULLs reduce space in tables
✔ **Answer: D) NULLs reduce space in tables**

Note: NULLs often waste space and add complexity.

5.2 MongoDB

Q170. MongoDB creates a unique index on which field during collection creation?

A) primary key
B) _id
C) 1st key of the collection
D) Cannot create unique index
✔ **Answer: B) _id field**

Q171. The maximum BSON document size in MongoDB is:

A) 14MB
B) 12MB

C) 16MB
D) Any size
✔ **Answer: C) 16MB**

Q172. Which MongoDB method inserts multiple documents?

A) insert()
B) insertOne()
C) insertMany()
D) All of the above
✔ **Answer: C) insertMany()**

Note: insertOne() for single, insertMany() for multiple documents.

Q173. How to rename a MongoDB collection?

A) db.collection.alter('new name')
B) db.collection.rename('new name')
C) db.collection.renameCollection('new name')
D) db.collection.renameCollection({'old name','new name'})
✔ **Answer: D) db.collection.renameCollection({'old name','new name'})**

Q174. Which method returns statistics about a MongoDB collection?

A) db.collection.stats()
B) db.collection.statistics()
C) db.collection.detail()
D) db.collection.info()
✔ **Answer: A) db.collection.stats()**

Q175. To add a column to a MongoDB collection, which method is used?

A) update()
B) alter()
C) add()
D) None of the above
✔ **Answer: D) None of the above**

Note: MongoDB is schema-less; add fields by updating documents with $set.

Q176. How to remove a document from a CAPPED collection in MongoDB?

A) db.collection.remove()
B) db.collection.remove({})
C) db.collection.removeAll({})
D) Documents from a capped collection cannot be removed
✔ **Answer: D) Documents from a capped collection CANNOT be removed**

Q177. Which technique distributes data across multiple machines in MongoDB?

A) Replication
B) Bucketing
C) Sharding
D) MongoDB does not allow this
✔ **Answer: C) Sharding**

Q178. The _____ operator matches documents with an array containing at least one element matching all specified criteria.

A) $elemMatch
B) $elementMatch
C) $ArrayMatch
D) None of the above
✔ **Answer: A) $elemMatch**

Q179. Which option of the update method creates a new document if none match?

A) multi:True

B) upsert:True
C) Collation
D) All of the above
✔ **Answer: B) upsert:True**

Q180. How to drop a database in MongoDB?

A) db.databasename.drop()
B) db.drop()
C) db.dropDatabase()
D) A database cannot be dropped
✔ **Answer: C) db.dropDatabase()**

Q181. Which MongoDB command finds employees with experience >= 5 and <= 10?

A) db.emp.find({experience:{$gt:5,$lt:10}})
B) db.emps.find({experience:{$gte:5,$lt:10}})
C) db.emp.find({experience:{$gt:5,$lte:10}})
D) db.emp.find({experience:{$gte:5,$lte:10}})
✔ **Answer: D) db.emp.find({experience:{$gte:5,$lte:10}})**

Q182. Which aggregate stage operator writes resulting documents to a new collection?

A) $push
B) $project
C) $out
D) $new
✔ **Answer: C) $out**

Q183. Which data model captures relationships by storing related data in a SINGLE document?

A) Embedded documents
B) Referenced document
C) Collage document
D) None of the above
✔ **Answer: A) Embedded documents**

Q184. In MongoDB, the right order of aggregate stage operators equivalent to SQL (WHERE, GROUP BY, HAVING) is:

A) $match, $group, $match
B) $group, $match, $having
C) $project, $match, $group
D) $match, $match, $group
✔ **Answer: A) $match, $group, $match**

Q185. Which MongoDB method returns collection statistics?

A) db.collection.stats()
B) db.collection.info()
C) db.collection.detail()
D) db.collection.statistics()
✔ **Answer: A) db.collection.stats()**

5.3 Cassandra

Q186. Cassandra is developed at:

A) Facebook
B) Yahoo
C) Twitter
D) Amazon
✔ **Answer: A) Facebook**

Q187. Cassandra is a _____ type of NoSQL database.

A) Document oriented

B) Graph oriented
C) Key-value pair
D) Column oriented
✔ **Answer: D) Column oriented NoSQL database**

Q188. In Cassandra, a _____ is a special column storing a map of sub-columns.

A) Key column
B) Sub column
C) Super column
D) Column
✔ **Answer: C) Super column**

Q189. Which of the following is/are components of Cassandra?

A) Data Centre
B) Cluster
C) Commit log
D) All of the above
✔ **Answer: D) All of the above**

Q190. _____ is an online NoSQL developed by Cloudera.

A) Cassandra
B) HBase
C) MongoDB
D) Hive
✔ **Answer: D) Hive**

SECTION 6: PL/SQL & STORED OBJECTS

6.1 PL/SQL Basics

Q191. PL/SQL is a _____ GL (Generation Language).

A) 1 GL
B) 2 GL
C) 3 GL
D) 4 GL
✔ **Answer: D) 4 GL**

Q192. Which section is COMPULSORY in a PL/SQL block?

A) Declaration (DECLARE)
B) Executable (BEGIN...END)
C) Exception (EXCEPTION)
D) All of the above
✔ **Answer: B) Executable section (BEGIN...END)**

Q193. An uninitialized variable in PL/SQL stores a _____ value.

A) Blank
B) 0
C) NULL
D) Garbage
✔ **Answer: C) NULL**

Q194. Comments in PL/SQL code are inserted using:

A) —
B) /* */
C) Both A and B
D) #
✔ **Answer: C) Both -- and /* */**

Q195. In PL/SQL, A _____ variable can store TRUE or FALSE.

A) Char
B) Binary
C) Boolean
D) Logical
✔ **Answer: C) Boolean**

Q196. The attribute to declare a variable based on a table column type is:

A) %ROWTYPE
B) %TYPE
C) %DATATYPE
D) %VARIABLE
✔ **Answer: B) %TYPE**

Note: e.g., v_sal emp.sal%TYPE;

Q197. To declare a composite variable with all columns of EMP table:

A) X EMP%tabletype
B) X EMP%rowtype
C) X EMP
D) EMP X
✔ **Answer: B) X EMP%rowtype**

Q198. In PL/SQL, the FOR loop variable:

A) Must be explicitly declared
B) Is implicitly created by Oracle
C) Must be initialized
D) Must be of INTEGER type
✔ **Answer: B) Is implicitly created and managed by Oracle**

Q199. Which loop ensures statements execute at least ONCE?

A) LOOP...END LOOP (simple loop)
B) WHILE loop
C) FOR loop
D) All of the above
✔ **Answer: A) LOOP...END LOOP (simple/infinite loop with EXIT WHEN)**

Q200. Identifiers in PL/SQL can contain up to _____ characters.

A) 255
B) 30
C) 1000
D) None
✔ **Answer: B) 30 characters**

6.2 Cursors

Q201. What is used to fetch data row by row?

A) Cursor
B) Variable
C) Both
D) None
✔ **Answer: A) Cursor**

Q202. The name of the implicit cursor created by Oracle in a PL/SQL block is:

A) Oracle
B) PLSQL
C) SQL
D) Cursor
✔ **Answer: C) SQL**

Q203. The type of cursor used for DML statements in PL/SQL is:

A) IMPLICIT
B) EXPLICIT
C) Both
D) Neither
✔ **Answer: A) IMPLICIT**

Q204. The type of cursor for SELECT statements returning MORE THAN ONE row is:

A) Explicit
B) Implicit
C) Both
D) Neither
✔ **Answer: A) Explicit**

Q205. The correct sequence for explicit cursor handling is:

A) Declare→Fetch→Open→Close
B) Declare→Open→Close→Fetch
C) Declare→Open→Fetch→Close
D) Open→Declare→Fetch→Close
✔ **Answer: C) Declare → Open → Fetch → Close**

Q206. The _____ step executes the SELECT query associated with an explicit cursor.

A) Declare
B) Open
C) Fetch
D) Close
✔ **Answer: B) Open**

Q207. Which explicit cursor attribute returns the NUMBER of rows fetched?

A) SQL%FOUND
B) SQL%ROW
C) SQL%ROWCOUNT
D) SQL%COUNT
✔ **Answer: C) SQL%ROWCOUNT**

Q208. The %notfound attribute indicates whether a FETCH returned a row.

A) True
B) False
C) Only for implicit cursors
D) Only for explicit cursors
✔ **Answer: A) True — it's TRUE when no row was fetched (useful for loop termination)**

Q209. Fetching PAST the last row of a cursor:

A) Raises NO_DATA_FOUND
B) Raises VALUE_ERROR
C) Raises CURSOR_NOT_FOUND
D) Does NOT raise an exception
✔ **Answer: D) Does not raise an exception**

Note: Only %NOTFOUND becomes TRUE; no exception is raised.

Q210. The NOWAIT option works for:

A) Implicit cursors only
B) Explicit cursors only
C) Both
D) Neither
✔ **Answer: B) Explicit cursors only**

6.3 Exceptions, Procedures & Triggers

Q211. RAISE_APPLICATION_ERROR() defines errors in the range:

A) -00000 to -99999
B) -01200 to -01299
C) -00030 to -00039
D) -20000 to -20999
✔ **Answer: D) -20000 to -20999**

Q212. SQLCODE returns:

A) SQLERRM
B) SQLERRCODE
C) SQLERR
D) SQLCODE
✔ **Answer: D) SQLCODE — numeric value of the error code**

Q213. SQLERRM returns:

A) Numeric error code
B) Message associated with error code
C) Table name
D) None
✔ **Answer: B) The message associated with the error code**

Q214. Which exception handler handles ALL unhandled exceptions?

A) WHEN ALL
B) WHEN OTHERS
C) WHEN ANY
D) WHEN EXCEPTION
✔ **Answer: B) WHEN OTHERS**

Q215. PRAGMA EXCEPTION_INIT is used to:

A) Declare an exception
B) Associate exception name with error number
C) Handle exceptions
D) Associate function with error number
✔ **Answer: B) Associate an exception name with an Oracle error number**

Q216. The default parameter type in stored procedures is:

A) IN
B) OUT
C) IN OUT
D) No default type
✔ **Answer: A) IN**

Q217. To call a stored procedure at SQL prompt: CREATE PROCEDURE aaa(num NUMBER) AS...

A) RUN aaa;
B) EXECUTE aaa;
C) EXECUTE aaa(56);
D) RUN aaa(56);
✔ **Answer: C) EXECUTE aaa(56);**

Q218. SQL%ROWCOUNT after UPDATE returns:

A) Boolean: success/failure
B) Numeric: number of rows updated
C) VARCHAR2: table name
D) LONG: all table data
✔ **Answer: B) A numeric value representing the number of rows updated**

Q219. Dropping a table has the following effect on a non-unique index:

A) No effect
B) The index will be DROPPED

C) Index is rendered invalid
D) Index contains NULLs
✔ **Answer: B) The index will be DROPPED**

Q220. When we combine multiple columns in a single index, it is known as a _____ index.

A) Composite
B) Multiple
C) Multicol
D) Complicated
✔ **Answer: A) Composite**

Q221. A trigger based on AFTER INSERT — the reference :old.column will:

A) Give previous value
B) Give current value
C) Give NULL
D) Invalid statement
✔ **Answer: D) Invalid statement**

Note: In an INSERT trigger, :old is not defined (there is no 'before' row).

Q222. INSTEAD OF trigger is used for operations on columns of:

A) Base tables
B) Materialized views
C) Non-key-preserved join views
D) Stored procedures
✔ **Answer: C) Non-key-preserved join views**

Q223. When a database trigger causes a second to fire which causes a third — this is called _____ triggers.

A) Cascading
B) Mutating
C) Constraining
D) Waterfall
✔ **Answer: A) Cascading**

Q224. Which trigger types are impacted by MUTATING TABLE constraints?

A) Row triggers only
B) Statement triggers only
C) Both row and statement
D) Neither
✔ **Answer: A) Row triggers only**

Q225. Which of the following database objects stores table column data and row reference info?

A) Tables
B) Sequences
C) Indexes
D) Views
✔ **Answer: C) Indexes**

SECTION 7: MIXED PRACTICE — CCEE LEVEL

Q226. Which subquery CANNOT be executed by itself as a separate statement?

A) Inner subquery
B) Correlated subquery
C) Scalar subquery
D) All of the above
✔ **Answer: B) Correlated subquery**

Note: A correlated subquery depends on the outer query for its values.

Q227. In a correlated query:

A) Inner query fires first
B) Outer query fires first
C) Inner query fires for every row returned by outer query
D) No order
✔ **Answer: C) Inner query fires for every row returned by outer query**

Q228. A subquery can be used to:

A) Create groups of data
B) Sort data in a specific order
C) Convert data to different format
D) Retrieve data based on an UNKNOWN condition
✔ **Answer: D) Retrieve data based on an unknown condition**

Q229. Which comparison operators can be used in outer query if subquery returns MULTIPLE rows?

A) IN and NOT IN
B) ANY/ALL
C) EXIST/NOT EXIST
D) All of the above options
✔ **Answer: D) All of the above options**

Q230. A relation is in 4NF if it is in BCNF and has no:

A) Partial dependencies
B) Transitive dependencies
C) Multi-valued dependencies
D) Functional dependencies
✔ **Answer: C) Multi-valued dependencies**

Q231. In MySQL, what is the difference between WHERE and HAVING?

A) They are identical
B) WHERE filters rows BEFORE grouping; HAVING filters AFTER
C) HAVING filters before WHERE
D) None of the above
✔ **Answer: B) WHERE filters rows before grouping; HAVING filters after group operations**

Q232. Which of the following SQL functions can operate on ANY datatype?

A) TO_CHAR
B) LOWER
C) LPAD
D) MAX
✔ **Answer: D) MAX**

Q233. SELECT ROUND(TRUNC(MOD(1600,10),-1),2) FROM dual; Result:

A) 0
B) 1
C) 0.00
D) Error
✔ **Answer: A) 0**

Note: MOD(1600,10)=0; TRUNC(0,-1)=0; ROUND(0,2)=0

Q234. Which clause is used to SELECT data from a range of values?

A) LIKE
B) IN
C) BETWEEN
D) EXISTS
✔ **Answer: C) BETWEEN**

Q235. The 4 phases of processing a cursor are:

A) Open, Fetch, Close, Execute
B) Declare, Open, Fetch, Close
C) Parse, Execute, Fetch, Close
D) Declare, Parse, Execute, Close
✔ **Answer: B) Declare, Open, Fetch, Close**

Q236. Packages improve performance by:

A) Loading multiple objects into RAM at once
B) Loading single object into library
C) Reducing network calls
D) None
✔ **Answer: A) Loading multiple objects into RAM at once**

Q237. Which of the following can be used inside SQL queries?

A) Stored Procedures
B) Stored Functions
C) Database Triggers
D) Views only
✔ **Answer: B) Stored Functions**

Q238. Entity integrity ensures that:

A) Every field corresponds to a related field
B) Every field value is within a range
C) Every ROW is unique
D) None of the above
✔ **Answer: C) Every row is unique (via Primary Key)**

Q239. How should a many-to-many relationship be handled in RDBMS?

A) By adding a join table
B) By adding an intersection table
C) By adding a union table
D) By adding a Cartesian table
✔ **Answer: B) By adding an intersection (junction/bridge) table**

Q240. Which SQL statement is equivalent to: SELECT NAME FROM CUSTOMER WHERE STATE='VA';

A) SELECT NAME IN CUSTOMER WHERE STATE IN ('VA');
B) SELECT NAME IN CUSTOMER WHERE STATE='VA';
C) SELECT NAME IN CUSTOMER WHERE STATE='V';
D) SELECT NAME FROM CUSTOMER WHERE STATE IN ('VA');
✔ **Answer: D) SELECT NAME FROM CUSTOMER WHERE STATE IN ('VA');**

Q241. CREATE TABLE new AS SELECT * FROM OLD WHERE 1=2 creates:

A) Copies all rows from OLD to NEW
B) Creates a dummy NEW table
C) Creates empty NEW table with same column definitions as OLD
D) Creates synonym for OLD
✔ **Answer: C) Creates an empty NEW table with the same column definitions as OLD**

Note: 1=2 is always false so no rows are copied, but structure is cloned.

Q242. SELECT DECODE(empid, 38475,'Terminated', 60494,'LOA', 'ACTIVE') FROM emp; If EMPID=60494:

A) 60494
B) LOA
C) Terminated
D) ACTIVE
✔ **Answer: B) LOA**

Q243. Which statement about Oracle's set operators is INCORRECT?

A) UNION — combines unique rows from two SELECT statements
B) UNION ALL — combines all rows including duplicates
C) INTERSECT — returns rows in BOTH SELECT statements
D) EXCEPT — returns rows in FIRST but not SECOND SELECT
✔ **Answer: D) EXCEPT — Oracle uses MINUS, not EXCEPT (though EXCEPT is ANSI standard)**

Q244. The default character for substitution variables in SELECT statements is:

A) Ampersand (&)
B) Ellipses
C) Quotation marks
D) Asterisk
✔ **Answer: A) Ampersand (&)**

Q245. Developer ANJU executes: CREATE TABLE animals AS SELECT * FROM MASTER.ANIMALS; Effect:

A) Table created in MASTER schema
B) Syntax error
C) Table created in ANJU schema with MASTER's data
D) Table named MASTER created in ANIMALS schema
✔ **Answer: C) Table named ANIMALS created in ANJU schema with same data as MASTER.ANIMALS**

Q246. When char datatype is converted to Varchar2 of same width:

A) Blank spaces are removed
B) Blank spaces remain
C) Conversion is not possible
D) None
✔ **Answer: B) All blank spaces associated with the column value will remain**

Q247. TNS in tnsnames.ora stands for:

A) Transparent Network Substrate
B) Transport Network Substrate
C) Transparent Network System
D) Transparent Network Server
✔ **Answer: A) Transparent Network Substrate**

Q248. The 5 important components of a DBMS are:

A) Database engine, schema, query processor, report writer, form generator
B) Database engine, data dictionary, query processor, report writer, form generator
C) Index, schema, DDL, DML, DCL
D) None of the above
✔ **Answer: B) Database engine, data dictionary, query processor, report writer, form generator**

Q249. Which view gives source code of stored procedures, functions, and packages?

A) USER_SOURCE
B) ALL_SOURCE
C) DBA_SOURCE
D) All of the above
✔ **Answer: D) All of the above**

Q250. Normalization is to _____ ; Denormalization is to _____.

A) Read performance / Write performance
B) Reduce redundancy / Improve read performance
C) Increase redundancy / Reduce redundancy

D) None of the above

✔ **Answer: B) Reduce redundancy (normalize) / Improve read (SELECT) performance (denormalize)**

QUICK ANSWER KEY SUMMARY

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	A	2	C	3	C	4	D	5	C
6	B	7	D	8	A	9	B	10	A
11	D	12	C	13	D	14	D	15	B
16	D	17	B	18	B	19	A	20	C
21	B	22	D	23	B	24	A	25	B
26	B	27	B	28	D	29	D	30	A
31	C	32	C	33	B	34	A	35	B
36	A	37	B	38	D	39	B	40	B
41	C	42	D	43	B	44	B	45	B
46	A	47	B	48	B	49	B	50	B
51	A	52	A	53	D	54	C	55	A
56	A	57	A	58	B	59	A	60	A
61	D	62	A	63	A	64	D	65	A
66	B	67	B	68	B	69	D	70	B
71	B	72	B	73	D	74	B	75	A
76	C	77	C	78	D	79	B	80	A
81	A	82	B	83	B	84	C	85	C
86	A	87	B	88	C	89	C	90	A
91	C	92	A	93	C	94	C	95	D
96	D	97	A	98	A	99	B	100	B
101	C	102	B	103	D	104	A	105	A
106	A	107	A	108	D	109	B	110	B
111	B	112	C	113	C	114	C	115	A
116	A	117	B	118	B	119	B	120	D
121	C	122	A	123	A	124	B	125	A
126	A	127	A	128	C	129	B	130	A
131	A	132	C	133	A	134	B	135	D
136	D	137	A	138	A	139	C	140	B
141	A	142	A	143	D	144	C	145	C
146	D	147	D	148	B	149	D	150	D
151	C	152	C	153	A	154	C	155	B
156	C	157	C	158	A	159	C	160	B
161	A	162	C	163	B	164	D	165	D
166	D	167	B	168	D	169	D	170	B
171	C	172	C	173	D	174	A	175	D
176	D	177	C	178	A	179	B	180	C
181	D	182	C	183	A	184	A	185	A

186	A	187	D	188	C	189	D	190	D
191	D	192	B	193	C	194	C	195	C
196	B	197	B	198	B	199	A	200	B
201	A	202	C	203	A	204	A	205	C
206	B	207	C	208	A	209	D	210	B
211	D	212	D	213	B	214	B	215	B
216	A	217	C	218	B	219	B	220	A
221	D	222	C	223	A	224	A	225	C
226	B	227	C	228	D	229	D	230	C
231	B	232	D	233	A	234	C	235	B
236	A	237	B	238	C	239	B	240	D
241	C	242	B	243	D	244	A	245	C
246	B	247	A	248	B	249	D	250	B